

NRE 3208 Project

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1 Reference Solution

The diffusion equation for a 1-D slab in vacuum is given below in Equation 1.

$$-3\frac{d^2\phi}{dx^2} + 2\phi = e^{-\frac{|x|}{10}} \quad (1)$$

1.1 Homogeneous Solution $\phi_h(x)$

The homogeneous solution is calculated by setting the left side of Equation 1 equal to 0, and takes the following form found in Equation 2.

$$\phi'' - \frac{2}{3}\phi = 0 \Rightarrow \phi_h(x) = Ae^{r_1x} + Be^{r_2x} \quad (2)$$

The roots to this equation are calculated below in Equation 3.

$$r^2 - \frac{2}{3} = 0 \Rightarrow r = \pm\sqrt{\frac{2}{3}} \quad (3)$$

Thus, the homogeneous solution takes the form of Equation 4.

$$\phi_x(x) = Ae^{\sqrt{\frac{2}{3}}x} + Be^{-\sqrt{\frac{2}{3}}x} \quad (4)$$

1.2 Particular Solution $\phi_p(x)$

The particular solution is taken from the right side of Equation 1, and its form and its derivatives can be seen below in Equation 5.

$$\phi_p = Ce^{-\frac{|x|}{10}} \Rightarrow \phi'_p = -\frac{C}{10}e^{-\frac{|x|}{10}} \Rightarrow \phi''_p = \frac{C}{100}e^{-\frac{|x|}{10}} \quad (5)$$

The scalar coefficient is solved for by plugging in the terms from Equation 5 into Equation 1, and is calculated below in Equation 6.

$$\begin{aligned} -3\left(\frac{C}{100}e^{-\frac{|x|}{10}}\right) + 2\left(Ce^{-\frac{|x|}{10}}\right) &= Ce^{-\frac{|x|}{10}} \\ C &= \frac{100}{197} \end{aligned} \quad (6)$$

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Thus, the particular solution takes the form of Equation 7.

$$\phi_p(x) = \frac{100}{197} e^{-\frac{|x|}{10}} \quad (7)$$

1.3 Analytical Solution $\phi(x)$

The analytical solution takes the form of Equation 8.

$$\phi(x) = \phi_h(x) + \phi_p(x) = Ae^{\sqrt{\frac{2}{3}}x} + Be^{-\sqrt{\frac{2}{3}}x} + \frac{100}{197} e^{-\frac{|x|}{10}} \quad (8)$$

Next we must calculate constants A and B from boundary conditions $\phi(\pm 10) = 0$ and $\phi'(0) = 0$. First we calculate the first derivative of ϕ in Equation 9.

$$\phi'(x) = \sqrt{\frac{2}{3}} Ae^{\sqrt{\frac{2}{3}}x} - \sqrt{\frac{2}{3}} Be^{-\sqrt{\frac{2}{3}}x} - \frac{10}{197} e^{-\frac{|x|}{10}} \quad (9)$$

Now we plug in $\phi(\pm 10) = 0$ into Equation 8 and $\phi'(0) = 0$ into Equation 9, and can calculate A and B by solving the system of equations in Equation 10.

$$\begin{aligned} \phi'(0) &= \sqrt{\frac{2}{3}} A - \sqrt{\frac{2}{3}} B - \frac{10}{197} = 0 \\ A &= \sqrt{\frac{2}{3}} \frac{10}{197} + B \\ \phi(10) &= Ae^{10\sqrt{\frac{2}{3}}} + Be^{-10\sqrt{\frac{2}{3}}} + \frac{100}{197e} = 0 \\ &= \left(\sqrt{\frac{2}{3}} \frac{10}{197} + B \right) e^{10\sqrt{\frac{2}{3}}} + Be^{-10\sqrt{\frac{2}{3}}} + \frac{100}{197e} = 0 \end{aligned} \quad (10)$$

Thus, the coefficients are solved for in Equation 11.

$$\begin{aligned} A &= \sqrt{\frac{2}{3}} \frac{10}{197} - \frac{\frac{100}{197e} + \sqrt{\frac{3}{2}} \frac{10}{197} e^{10\sqrt{\frac{2}{3}}}}{e^{10\sqrt{\frac{2}{3}}} + e^{-10\sqrt{\frac{2}{3}}}} \approx -5.3113 \times 10^{-5} \\ B &= -\frac{\frac{100}{197e} + \sqrt{\frac{3}{2}} \frac{10}{197} e^{10\sqrt{\frac{2}{3}}}}{e^{10\sqrt{\frac{2}{3}}} + e^{-10\sqrt{\frac{2}{3}}}} \approx -6.2223 \times 10^{-2} \end{aligned} \quad (11)$$

Thus, the final analytical solution is described below in Equation 12. Note that due to the boundary condition $\phi'(0) = 0$ and hence $A \neq B$, we change $e^x \Rightarrow e^{|x|}$ in order to reflect flux symmetry about the y-axis.

$$\phi(x) = \left(\sqrt{\frac{2}{3}} \frac{10}{197} - \frac{\frac{100}{197e} + \sqrt{\frac{3}{2}} \frac{10}{197} e^{10\sqrt{\frac{2}{3}}}}{e^{10\sqrt{\frac{2}{3}}} + e^{-10\sqrt{\frac{2}{3}}}} \right) e^{\sqrt{\frac{2}{3}}|x|} + \left(-\frac{\frac{100}{197e} + \sqrt{\frac{3}{2}} \frac{10}{197} e^{10\sqrt{\frac{2}{3}}}}{e^{10\sqrt{\frac{2}{3}}} + e^{-10\sqrt{\frac{2}{3}}}} \right) e^{-\sqrt{\frac{2}{3}}|x|} + \frac{100}{197} e^{-\frac{|x|}{10}} \quad (12)$$

2 Derivations for S_n and A_n

We start by stating the Hemholtz Equation in Equation 13.

$$\begin{aligned}\nabla^2 \Psi + B^2 \Psi &= 0 \\ \Psi &= A \cos Bx + C \sin Bx \\ \Psi_n &= A \cos B_n x, \quad B_n = \frac{n\pi}{a}, \quad n = 1, 3, 5, \dots\end{aligned}\tag{13}$$

Applying the boundary condition of $\Psi_n(10) = 0$, we get Equation 14.

$$\begin{aligned}\Psi_n(10) &= 0 \\ \cos(B_n * 10) &= 0 \Rightarrow B_n = \frac{n\pi}{20}, \quad n = 1, 3, 5, \dots \\ \Psi_n(x) &= \cos\left(\frac{n\pi x}{20}\right)\end{aligned}\tag{14}$$

Taking the general Fourier series $f(x) = \sum_n^\infty C_n \Psi_n(x)$, multiplying both sides by $\Psi_M(x)$, and integrating over $[-\frac{a}{2}, \frac{a}{2}]$, we arrive at Equation 15 which allows us to calculate our expansion coefficients.

$$\begin{aligned}\int_{-\frac{a}{2}}^{\frac{a}{2}} f(x) \Psi_M(x) dx &= \int_{-\frac{a}{2}}^{\frac{a}{2}} \sum_{n, \text{ odd}}^\infty C_n \Psi_n \Psi_M dx = \sum_{n, \text{ odd}}^\infty \int_{-\frac{a}{2}}^{\frac{a}{2}} \Psi_n(x) \Psi_M(x) dx \\ &= \begin{cases} 0, & n \neq m \\ C_M \int_{-\frac{a}{2}}^{\frac{a}{2}} \Psi_M^2 dx = \int_{-\frac{a}{2}}^{\frac{a}{2}} \cos^2\left(\frac{M\pi x}{a}\right) dx = C_M \frac{a}{2}, & n = m \end{cases} \\ \therefore C_M &= \frac{2}{a} \int_{-\frac{a}{2}}^{\frac{a}{2}} f(x) \Psi_M(x) dx\end{aligned}\tag{15}$$

Next we expand our flux and source distributions as an expansion of eigenfunctions $\Psi_n(x)$, as seen in Equation 16.

$$\begin{aligned}\phi(x) &= \sum_{n, \text{ odd}}^\infty A_n \Psi_n(x) \\ S(x) &= \sum_{n, \text{ odd}}^\infty S_n \Psi_n(x)\end{aligned}\tag{16}$$

The general solutions for S_n and A_n are found by taking the below diffusion equation, substituting our expansions from Equation 16, and solving for $\frac{d^2 \phi}{dx^2} = -B\phi$ from the Hemholtz Equation (Equation 13), which can all be seen in Equation 17.

$$\begin{aligned}\frac{d^2 \Psi}{dx^2} - \frac{1}{L^2} \Psi(x) &= -\frac{S(x)}{D} \\ \sum_{n, \text{ odd}}^\infty A_n \left[B_n^2 + \frac{1}{L^2} \right] \Psi_n &= \sum_{n, \text{ odd}}^\infty \frac{1}{D} S_n \Psi_n\end{aligned}\tag{17}$$

Thus, the general solutions for A_n and S_n are listed below in Equation 18.

$$\begin{aligned}A_n &= \frac{S_n}{D \left[B_n^2 + \frac{1}{L^2} \right]} \\ S_n &= \frac{2}{a} \int_{-\frac{a}{2}}^{\frac{a}{2}} S(x) \Psi_n(x) dx\end{aligned}\tag{18}$$

Mapping Equation 1 onto the Diffusion Equation, we can calculate the values for D , L , and $S(x)$ for this example in Equation 19.

$$\begin{aligned}\frac{d^2\phi}{dx^2} - \frac{1}{L^2}\phi(x) &= -\frac{S(x)}{D} \\ \phi'' - \frac{2}{3}\phi &= -\frac{e^{-\frac{|x|}{10}}}{3} \\ D = 3, L = \sqrt{\frac{3}{2}}, S(x) &= e^{-\frac{|x|}{10}}\end{aligned}\tag{19}$$

Plugging in the values from Equation 19 into the general solution from Equation 18, the expansion coefficients are listed below in Equation 20. Notice how due to symmetry, we can change the integration bounds from $\int_{-\frac{a}{2}}^{\frac{a}{2}} dx$ to $2 \int_0^{\frac{a}{2}} dx$.

$$\begin{aligned}A_n &= \frac{S_n}{3B_n^2 + 2} \\ S_n &= \frac{2}{10} \int_{-\frac{a}{2}}^{\frac{a}{2}} e^{-\frac{|x|}{10}} \cos(B_n(x)) dx = 2 * \frac{2}{10} \left[\frac{e^{-\frac{|x|}{10}}}{\frac{1}{100} + B_n^2} \left(-\frac{1}{10} \cos(B_n x) + B_n \sin(B_n x) \right) \right]_0^{10} \\ &= \frac{2}{5} \left[\frac{e^{-1}}{\frac{1}{100} + B_n^2} \left(0 + B_n \sin\left(\frac{n\pi}{2}\right) \right) \right] - \frac{2}{5} \left[\frac{1}{\frac{1}{100} + B_n^2} \left(-\frac{1}{10} + 0 \right) \right] \\ S_n &= \frac{40}{1 + 100B_n^2} \left(\frac{B_n \sin\left(\frac{n\pi}{2}\right)}{e} + \frac{1}{10} \right)\end{aligned}\tag{20}$$

3 Numeric RMS Error

The error is a function of the variation of the solution for $\phi_{numeric}(x_i)$ between the current mode N and the previous mode $N - 1$. Note that these N s are whole numbers that count the number of odd n modes given that $n = (2N - 1)$. The RMS error is a function of error, as shown in Equation 21. Note that $\mathbb{E}[x]$ is the expected value, or mean, of x .

$$\begin{aligned}E(x_i) &= 1 - \frac{\phi_N(x_i)}{\phi_{N-1}(x_i)} \\ RMS &= \sqrt{\frac{1}{N} \sum_i E^2(x_i)} = \sqrt{\mathbb{E}[E^2]}\end{aligned}\tag{21}$$

Figure 1 plots RMS error versus the number of modes.

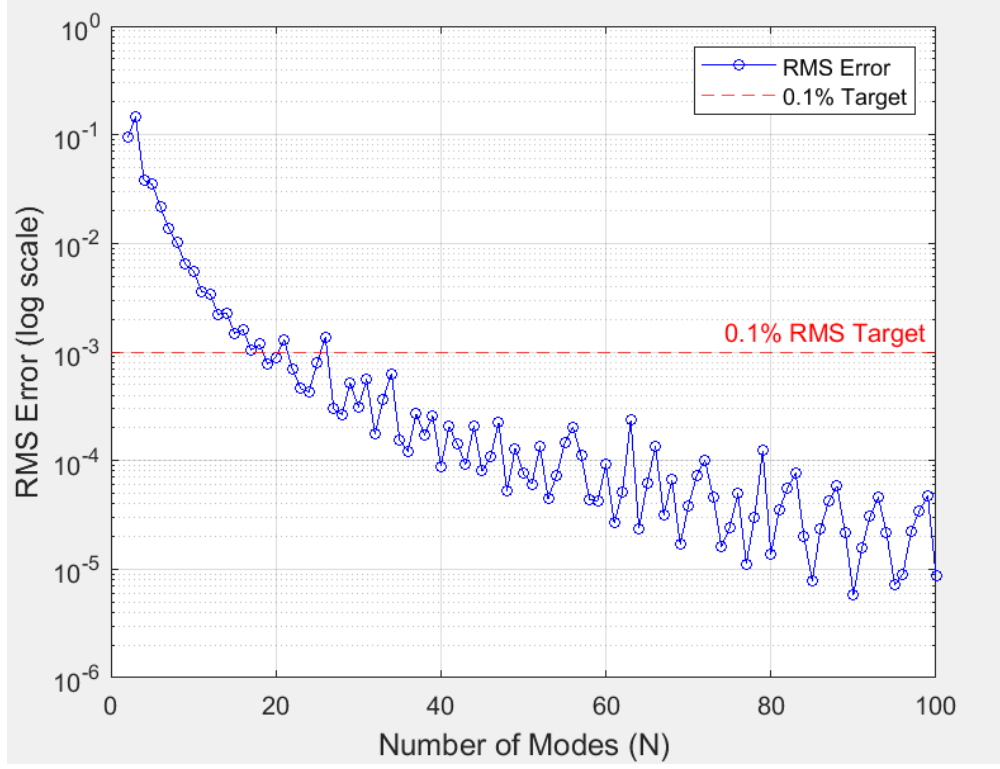


Figure 1: RMS Error vs. Number of Modes

4 Contributions from $n = \{3, 5, 7, \text{Final Converged}\}$ Modes

Figure 2 plots the cumulative flux distribution based on the highest-order mode. In the case of Figure 2, $n = 3$ includes the $n = 1, 3$ modes, $n = 5$ is the sum of the $n = 3$ plot and the $n = 5$ mode, and so on.

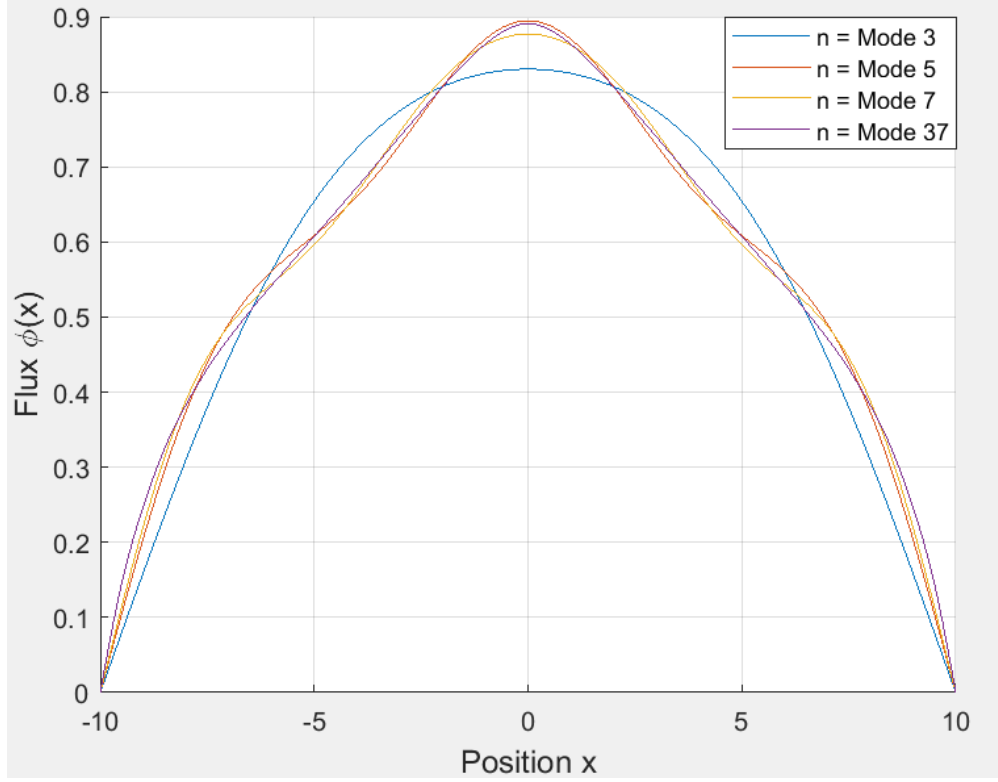


Figure 2: Cumulative Flux Contributions

5 Normalized Analytic and Numeric Solutions

Figure 3 plots the analytical solution from Section 1.3 versus the numerical solution from Equation 16 with the expansion coefficients from Equation 20. The shapes of the two distributions are essentially identical. Note that $N = 19$ refers to the actual number of harmonic terms added to the numerical solution as calculated in Section 3, as opposed to the corresponding highest-order $n = 37$ mode which is referenced in the legend of Figure 2. Normalization is performed according to Equation 22.

$$\phi_{shape} = \frac{\phi(x)}{\text{sum}(\phi(x))} \quad (22)$$

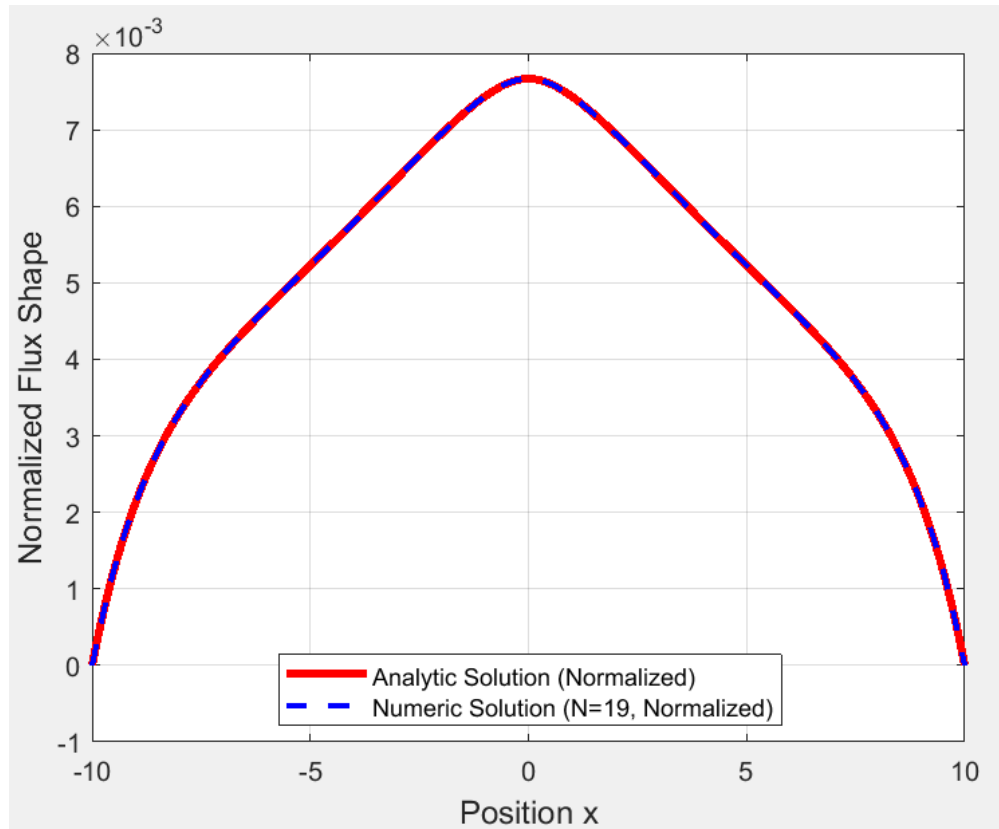


Figure 3: Normalized Analytic vs. Converged Numeric Shape